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1. Age of child and photograph preference.

Sol: Given data

age of child	A	B	C	Total
5-6 years	18	22	20	60
7-8 years	2	28	40	70
9-10 years	20	10	40	70
Total	40	60	100	200

H_0 : Age and photograph

H_1 : Age and photograph preference are significantly related

R-program :-

```
data <- matrix (c(18, 22, 20,
                  2, 28, 40
                  20, 10, 40),
                  nrow = 3, byrow = True).

colnames(data) <- c("A", "B", "C")
  rownames (data) <-
    c("5-6", "7-8", "9-10")
    data
     chisq. test (data)
```

Output :.

x - squared = 29.603   p-value = 5.89e-0.6
df = 4

covariance b/w B and C

B = 22 , 28, 10

C = 20, 40, 40

R code :-

```
B <- c(22,8,10)

C <- c(20,40,40)

  cov(B,C)
```

<u>output</u>

```
  [1] -20
```

sample covariance = -20.

2. Given data

player	points	player	points
P1	12	P9	11
P2	15	P10	35
P3	14	P11	16
P4	10	P12	17
P5	18	P13	19
P6	20	P14	21
P7	22	P15	23
P8	13		

Box Plot Values

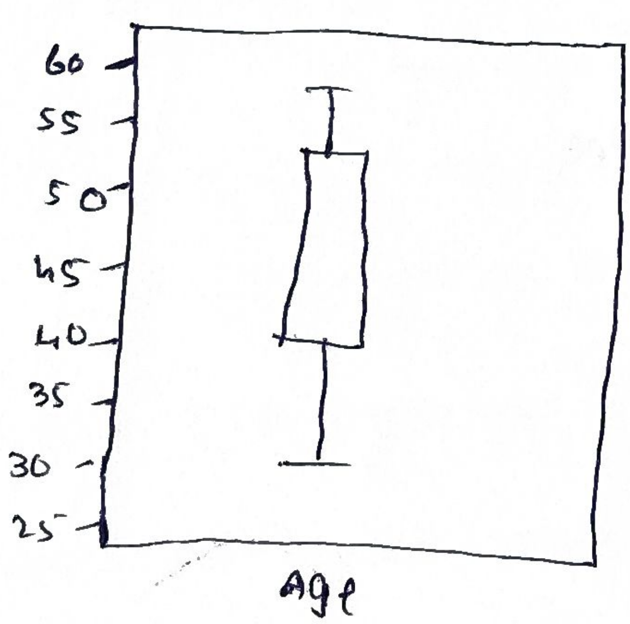
Age :-

- minimum = 23
- Q_1 = 39.5
- Median = 51
- Q_3 = 56.75
- Maximum = 61

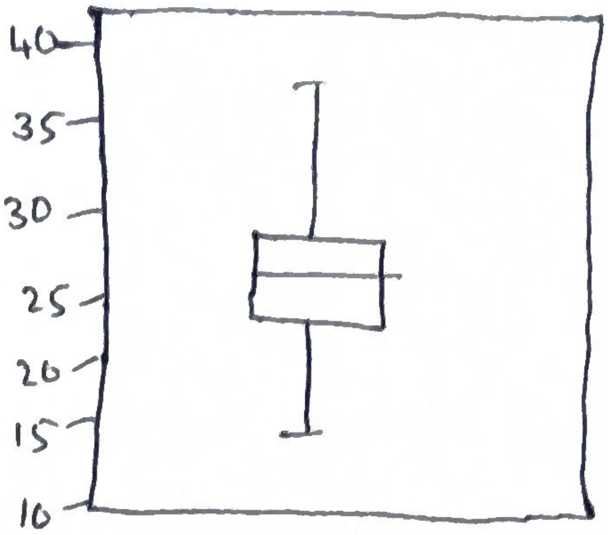
% fat

- minimum = 7.8
- Q_1 = 26.675
- Median = 30.7
- Q_3 = 33.925
- Maximum = 42.5

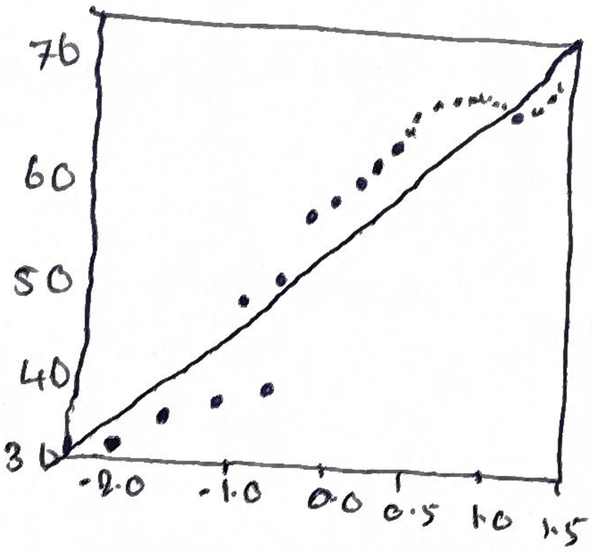
Box Plot of Age



Boxplot of Body fat percentage



Q-Q Plot of Age



Theoretical Quantities,

$Q_1 = 13$

$Q_3 = 21$

$IQR = Q_3 - Q_1$

$IQR = 21 - 13 = 18$

upper limit

$Q3 + 1.5 \times IQR$

$= 21 + 1.5(8)$
$= 33$

3) R- program

```
Install. packages ("rpart")
Install. Packeges ("rpart.plot")
library (r part)
library (r.part. plot).
```

Build Decision tree

```
Model <- rpart (loan ~ Age + Income + Employement + creditscore,
                data= data , method ="class")
print(model).
```

Plot the tree

```
r.part. plot (model,
             type=2,
             extra= 104,
             fallen . leaves = True).
```

predict loan approval

```
Pred <- predict (model, data, type="class")
pred.
```

confusion matrix

table (Actual = data loan,

predicated = pred

4. class A

78, 35, 47, 64, 95, 66, 89, 36, 84

class B

51, 56, 84, 60, 59, 70, 63, 66, 50.

i) Mean, Median & Range

class A

$\frac{592}{9}$ = 65.78 ← Mean

Median = 66

Range

95 - 35 = 60

class B

$\frac{559}{9}$ = 62011

Median = 60

Range

84 - 50 = 34

ii) Box Plot

```
class A <-
  c(76,35,47,64,95,66,89,36,84)

class B <-
  c(51,56,84,60,59,70,63,66,50)

boxplot(classA, classB,
        names = c("classA", "class B"),
        main ="Box Plot of maths Exam scores",
          ylab = "Marks"),
```

5. Given data

Age	%fat
23	9.5
23	26.5
27	7.8
27	17.8
39	31.4
41	25.9
47	27.4
49	27.2
50	31.2
52	34.6
54	42.5

54	28.8
56	33.4
57	30.2
58	34.1
58	32.9
60	41.2
61	35.7.

Age:-

mean = 46.44

Median = 51

standard deviation = 13.22

%fat

- Mean = 28.78
- Median = 30.70
- standard deviation = 9.25